

GCE A LEVEL MARKING SCHEME

PREDICTED – HARD

A LEVEL

ECONOMICS – COMPONENT 1

A520U10-1 (Predicted – Hard)

About this marking scheme

The purpose of this marking scheme is to provide teachers, learners and other interested parties with an understanding of the assessment criteria used to assess this specific assessment.

A list of indicative content suggests the range of economic concepts, theory, issues and arguments which might be included in learners' answers. It is not intended to be exhaustive and learners do not have to include all of the indicative content to reach the highest level of the mark scheme.

The level-based mark schemes sub-divide the total mark to allocate to individual assessment objectives. A mark will be awarded for each assessment objective targeted in the question, then totalled to give an overall mark for the question.

GENERAL MARKING GUIDANCE

Positive Marking

It should be remembered that learners are writing under examination conditions and credit should be given for what the learner writes, rather than adopting the approach of penalising for any omissions.

SECTION A

Q	Answer	AO
1	A	AO1
2	C	AO2
3	C	AO2
4	B	AO2
5	C	AO1
6	B	AO2
7	D	AO1
8	B	AO2
9	C	AO2
10	C	AO1
11	C	AO1
12	A	AO2
13	D	AO2
14	B	AO2
15	C	AO1
16	A	AO2
17	C	AO1
18	A	AO1
19	B	AO2
20	C	AO1

AO1: 8**AO2: 12**

SECTION B**21. Using the data, explain the likely link between the two charts. [4]**

Band	AO1	AO2	AO3
2			2 marks Good analysis A clear chain of reasoning links higher Fed rates to dollar appreciation via capital flows.
1	1 mark Limited knowledge Understanding of the link between interest rates and exchange rates.	1 mark Limited application The data is used in the answer.	1 mark Limited analysis A chain of reasoning exists but is underdeveloped.
0	0 marks	0 marks	0 marks

Indicative content**AO1**

Interest rate parity theory suggests that capital will flow towards currencies offering higher risk-adjusted returns. Higher US interest rates relative to the Eurozone raise the returns on dollar-denominated assets, increasing demand for dollars (and supply of euros) on the foreign exchange market.

AO2

The data shows the Fed funds rate rising from 0.25% to 5.5% between early 2022 and mid-2024. Over the same period, the USD appreciated against the EUR (the EUR/USD rate fell from 1.13 to around 1.05 before partially recovering and then falling again to 1.07 in mid-2024).

AO3

Chain: aggressive Fed tightening raises the interest rate differential between the US and the Eurozone → hot money / portfolio capital flows into dollar assets → demand curve for dollars shifts right / supply of dollars shifts left → dollar appreciates against the euro. The partial recovery of the euro in 2023 is consistent with the ECB catching up in its own tightening cycle, narrowing the differential.

Allow other relevant points.

22. Using the diagram, discuss whether a carbon tax is the best policy response to industrial pollution. [7]

Band	AO2	AO3	AO4
3			3 marks Excellent evaluation Clear supported judgement about whether a carbon tax is the best policy, considering information failures and alternatives.
2	2 marks Good application The diagram is used effectively to support the argument.	2 marks Good analysis Well-developed chain of reasoning about how a carbon tax internalises the externality.	2 marks Good evaluation Clear counterarguments showing the limitations of a carbon tax and alternative policies.
1	1 mark Limited application The diagram is referenced superficially.	1 mark Limited analysis The chain of reasoning is underdeveloped.	1 mark Limited evaluation Counterarguments present but not developed.
0	0 marks	0 marks	0 marks

Indicative content**AO2**

The diagram shows a market with $MSC > MPC$, reflecting the negative production externality (emissions, air quality damage, climate costs). Free market equilibrium Q_m exceeds the social optimum Q_s . A welfare loss triangle lies between MSC and MSB from Q_s to Q_m . A carbon tax equal to the marginal external cost at Q_s would shift the MPC curve up to $MPC + t = MSC$, internalising the externality and restoring the socially optimum output.

AO3

A Pigouvian (carbon) tax aligns private incentives with social costs: firms face the full cost of their emissions, so profit-maximising decisions produce at the social optimum. The tax raises revenue that can fund green investment or offset distortionary taxes (double dividend). It preserves market incentives for cost-minimising abatement – firms choose the cheapest way to reduce emissions, so the policy is productively efficient relative to command-and-control regulation.

AO4

Counterarguments: (i) Setting the 'correct' tax rate is extremely difficult due to uncertainty over the marginal damage – information failure. (ii) PED for polluting goods may be very inelastic in the short run so a tax has little effect on quantity, only raising prices. (iii) Regressive impact – lower-income households spend a higher share of income on energy, creating equity concerns unless revenue is recycled. (iv) Carbon leakage – domestic firms may relocate production to jurisdictions without a carbon tax, reducing competitiveness and failing to cut global emissions. Carbon border adjustment mechanisms can mitigate this but are administratively complex. (v) Alternatives: tradable emissions permits (cap-and-trade, e.g. EU ETS) fix the quantity of emissions at the socially optimal level, letting price adjust – arguably better when the total emissions target is known. Direct regulation (emissions standards), subsidies for green alternatives, information campaigns, and investment in R&D and infrastructure are also options. (vi) Political economy – tax changes are politically

contentious and may be reversed. The 'best' policy likely involves a mix: a carbon price as the backbone, complemented by R&D support and regulations for harder-to-abate sectors.

Allow other relevant points.

23. Use the data to explain where the Nash equilibrium lies in this matrix and why collusion between the firms might be difficult to sustain. [4]

Band	AO1 (1 mark)	AO2 (2 marks)	AO3 (1 mark)
2		2 marks Good application The matrix is used to show the Nash equilibrium and why collusion is unstable (cheating incentive).	
1	1 mark Limited knowledge Understanding of Nash equilibrium and/or collusion.	1 mark Limited application The matrix data is used to identify the equilibrium.	1 mark Limited analysis A clear chain of reasoning about how the equilibrium is reached or why collusion fails.
0	0 marks	0 marks	0 marks

Indicative content

AO1

A Nash equilibrium is a combination of strategies where no player can gain by unilaterally deviating. Tacit or explicit collusion would allow both firms to charge a high price, but this is unstable because each firm has a unilateral incentive to cheat.

AO2 / AO3

Testing each cell: if SouthFly sets High, NorthAir compares £80m (High) and £110m (Low) – chooses Low. If SouthFly sets Low, NorthAir compares £30m (High) and £50m (Low) – chooses Low. Low is NorthAir's dominant strategy. By symmetry, Low is also SouthFly's dominant strategy. The Nash equilibrium is (Low, Low) → £50m each.

The collusive outcome (High, High) would give £80m each – mutually better than £50m. But it is not a Nash equilibrium because each firm can unilaterally defect from High to Low and earn £110m instead of £80m. This is the classic prisoners' dilemma. Collusion is difficult to sustain because (i) the gain from cheating in a single round is large; (ii) detecting cheating quickly is difficult; (iii) legal prohibitions on price-fixing make communication illegal; (iv) entry of new firms or demand shocks destabilise any agreement. Only with repeated interactions and credible punishment strategies (e.g. tit-for-tat) might cooperation be sustained.

Allow other relevant points.

24. Using the data, discuss whether the Phillips curve remains a useful framework for understanding the inflation-unemployment relationship in the UK. [8]

Band	AO2	AO3	AO4
3		3 marks Excellent analysis Developed chains covering short-run and long-run Phillips curves, role of expectations and supply shocks.	3 marks Excellent evaluation Supported judgement about the usefulness of the framework and its limitations.
2	2 marks Good application Data is used effectively to support analysis/evaluation.	2 marks Good analysis Developed chain of reasoning about the Phillips curve relationship.	2 marks Good evaluation Clear counterarguments challenging the simple Phillips curve relationship.
1	1 mark Limited application.	1 mark Limited analysis.	1 mark Limited evaluation.
0	0 marks	0 marks	0 marks

Indicative content

AO2

The data shows an atypical pattern: inflation spiking to 9.1% in 2022 while unemployment fell to 3.7% (simple Phillips relationship holds), but unemployment remained low (4.0%) even as inflation fell sharply back to 7.3% in 2023 and 2.5% in 2024. The extract highlights exogenous supply shocks (energy prices, supply bottlenecks) as the main driver of 2022 inflation.

AO3

The original Phillips curve (1958) identified an empirical inverse relationship between wage/price inflation and unemployment – low unemployment implies tight labour markets and rising wages, passed through to prices. This is the 'short-run Phillips curve' (SRPC). In 2021–22, the UK fits this pattern to some extent – tight labour markets, high vacancy-to-unemployment ratio contributed to wage growth. The expectations-augmented Phillips curve (Friedman/Phelps) recognises that workers and firms adjust expectations, so the SRPC shifts over time. The long-run Phillips curve (LRPC) is vertical at the NAIRU – there is no permanent trade-off.

AO4

Counterarguments / evaluation: (i) Supply shocks can shift the SRPC, producing stagflation – simultaneous rising inflation and rising unemployment, or falling inflation with flat unemployment, which a naive Phillips curve cannot explain. The 2022–24 UK data arguably reflects this: most of the inflation was externally driven (energy, post-Covid supply chains, £/\$ depreciation) rather than demand-pull. (ii) Anchored inflation expectations (from 25 years of BoE inflation targeting) may mean that expectations did not de-anchor, avoiding a wage-price spiral and allowing inflation to fall without a significant rise in unemployment. (iii) Structural changes in labour markets – gig economy, weaker unions – may have flattened the SRPC, so even tight markets do not produce strong wage inflation. (iv) Inflation is multi-causal: exchange rate, commodity prices, productivity, expectations all matter. (v) Still, the Phillips framework – particularly the expectations-augmented version – remains useful for understanding why monetary policy must respond to prevent second-round effects. Overall: the framework retains analytical value, especially in augmented form, but the 2022–24 data highlight its limitations in the face of large supply shocks.

Allow other relevant points.

25. Discuss whether the HDI alone is sufficient to conclude that Saudi Arabia has a higher level of development than Costa Rica. [8]

Band	AO2	AO3	AO4
3		3 marks Excellent analysis Well-developed chain explaining how different development indicators point to different conclusions.	3 marks Excellent evaluation Supported judgement that the HDI alone is insufficient, using multiple dimensions of development.
2	2 marks Good application Data used effectively on both sides.	2 marks Good analysis Developed chain of reasoning about HDI strengths.	2 marks Good evaluation Developed counterarguments about HDI's limitations.
1	1 mark Limited application.	1 mark Limited analysis.	1 mark Limited evaluation.
0	0 marks	0 marks	0 marks

Indicative content

AO2

HDI suggests Saudi Arabia is more developed:

HDI 0.875 vs 0.806 – substantial margin (Very High vs Very High, but Saudi Arabia is well inside the Very High band). GNI/capita (PPP\$) is more than double (\$54,992 vs \$23,909). Longer life expectancy (78.8 vs 77.3) and more mean years of schooling (11.3 vs 8.8).

Broader indicators suggest Costa Rica may be more developed:

Gender Inequality Index: Costa Rica 0.256 (much lower, i.e. more gender equal) vs Saudi Arabia 0.569 – Saudi Arabia has very large gender gaps in labour participation, health, empowerment. Gini coefficient: slightly lower in Saudi Arabia (0.459) but both moderately unequal; however Saudi Arabia's distributional figures mask the precarious position of migrant workers. Costa Rica has stronger democratic institutions, no standing army, and historically high investment in health and education.

AO3

Development involves more than income, health, and education. Amartya Sen's capabilities approach emphasises the real freedoms people have to achieve valuable outcomes. HDI captures only three dimensions, and each is imperfectly measured: GNI includes oil rents (a large share of Saudi GDP) which may not translate into broad-based welfare; life expectancy does not capture quality of life or morbidity; schooling years do not measure learning outcomes.

AO4

Counterarguments: (i) HDI is a composite index chosen for its data availability and simplicity – it is a starting point, not a full picture. (ii) Inequality-adjusted HDI (IHDI) applies discounts for inequality in each dimension and typically cuts Saudi Arabia's score more than Costa Rica's. (iii) Environmental sustainability – Costa Rica is a global leader (>99% renewable electricity, high biodiversity protection) whereas Saudi Arabia is a major hydrocarbon economy. (iv) Freedoms – political, civil, gender – are highly unequal. Costa Rica scores strongly on press freedom and democracy indices. (v) Conversely, Saudi Arabia has significantly better physical infrastructure and healthcare systems in some dimensions. (vi) Other metrics: MPI (Multidimensional Poverty Index), Happy Planet Index,

Social Progress Index would give fuller pictures. Overall judgement: HDI alone is not sufficient; when the broader picture (inequality, gender, environment, freedoms) is taken into account, the claim that Saudi Arabia is 'more developed' becomes much more contestable.

Allow other relevant points.

26. Using supply and demand diagram(s), discuss whether the data suggests that oil prices are more likely to rise or fall in the future. [9]

Band	AO1	AO2	AO3 / AO4
3			3 marks (AO4) Well-reasoned supported judgement about the direction of oil prices, accounting for short vs long run.
2	2 marks Accurate S&D diagrams illustrating both sides.	2 marks Effective use of data on both sides.	2 marks (AO3) Developed analysis of one side. 2 marks (AO4) Developed counterargument.
1	1 mark Diagrams with inaccuracies or one side only.	1 mark Data used on one side only.	1 mark (AO3) / 1 mark (AO4) Chain of reasoning underdeveloped.
0	0 marks	0 marks	0 marks

Indicative content

AO1

Two diagrams: Rising prices – S shifts left (OPEC+ cuts) and/or D shifts right (EM demand), giving higher P and lower Q. Falling prices – S shifts right (US shale, EV transition reducing future demand) and/or D shifts left (recession, EV adoption), giving lower P and higher Q.

AO2

UPWARD price pressures:

OPEC+ voluntary supply cuts of 1 mb/d shift supply curve left. Geopolitical tensions in the Middle East add risk premium and threaten disruption of a major producing region. Strong demand growth from India and Southeast Asia as incomes rise (YED > 0 for oil in industrialising economies).

DOWNWARD price pressures:

US shale production at record 13.2 mb/d expanding supply, partially offsetting OPEC+ cuts. Accelerating EV adoption (17% of new car sales) and IEA forecast of demand peaking by 2030 – a structural negative demand shock. Potential recession in advanced economies reduces industrial demand.

AO3

Developed chain (downward pressure): EV substitution reduces derived demand for oil as transport fuel → demand curve for oil shifts left → at current price there is excess supply → producers cut prices to clear the market. Amplified by secular policies (carbon pricing, EV subsidies, infrastructure build-out). Additionally, US shale is very responsive (higher PES than conventional oil) so supply shifts further right as technology improves.

AO4

Counter (upward pressure): OPEC+ has historically defended price floors through coordinated cuts, and Saudi Arabia's 1 mb/d voluntary reduction plus the Middle East risk premium can be substantial. Oil demand remains inelastic in the short run – even with EVs expanding, total oil demand is still growing, and emerging market growth outweighs OECD decline for several more years. Timing considerations: recession and EV adoption may dominate in the medium term, but short-run OPEC+ action can produce spikes. The chart shows that oil has traded in a wide range

(\$75–\$123) over the past two years – volatility is the norm, making a one-directional forecast unreliable. PES is low in the short run (wells take years to drill) which amplifies price swings from demand shocks. Overall: short-run prices likely volatile with upward skew from OPEC+/geopolitics, but long-run structural decline as electrification gathers pace. Judgement must recognise this two-horizon picture.

Allow other relevant points.

AO Grid

	AO1	AO2	AO3	AO4	Total
Q1–20	8	12			20
Q21	1	1	2		4
Q22		2	2	3	7
Q23	1	2	1		4
Q24		2	3	3	8
Q25		2	3	3	8
Q26	2	2	2	3	9
Total	12	23	13	12	60